

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5115

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 2 Total Marks:50 Annexure A

CONCEPT MAPPING

Code of Conduct:

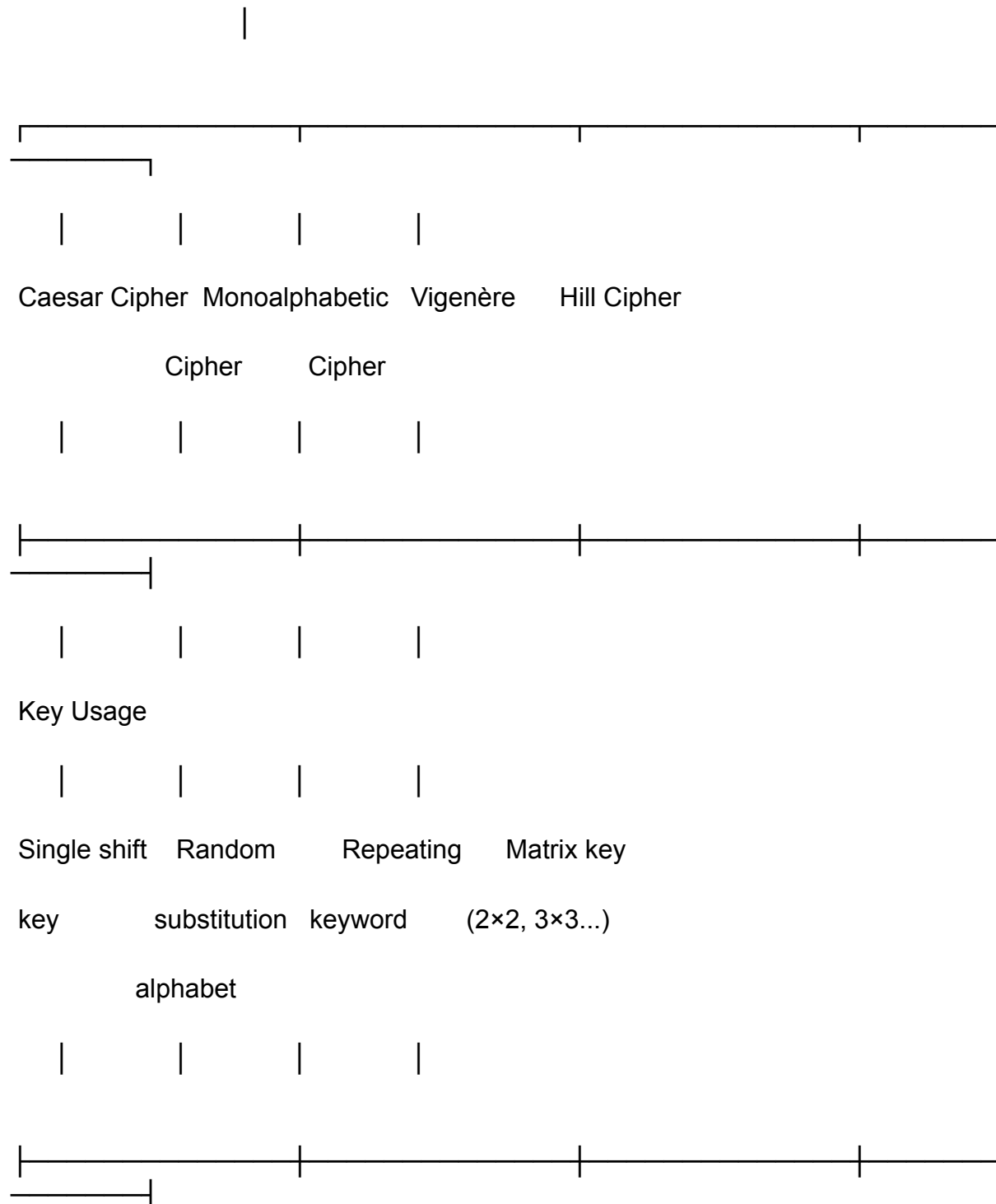
I _____meghanadh ambati (192372281)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

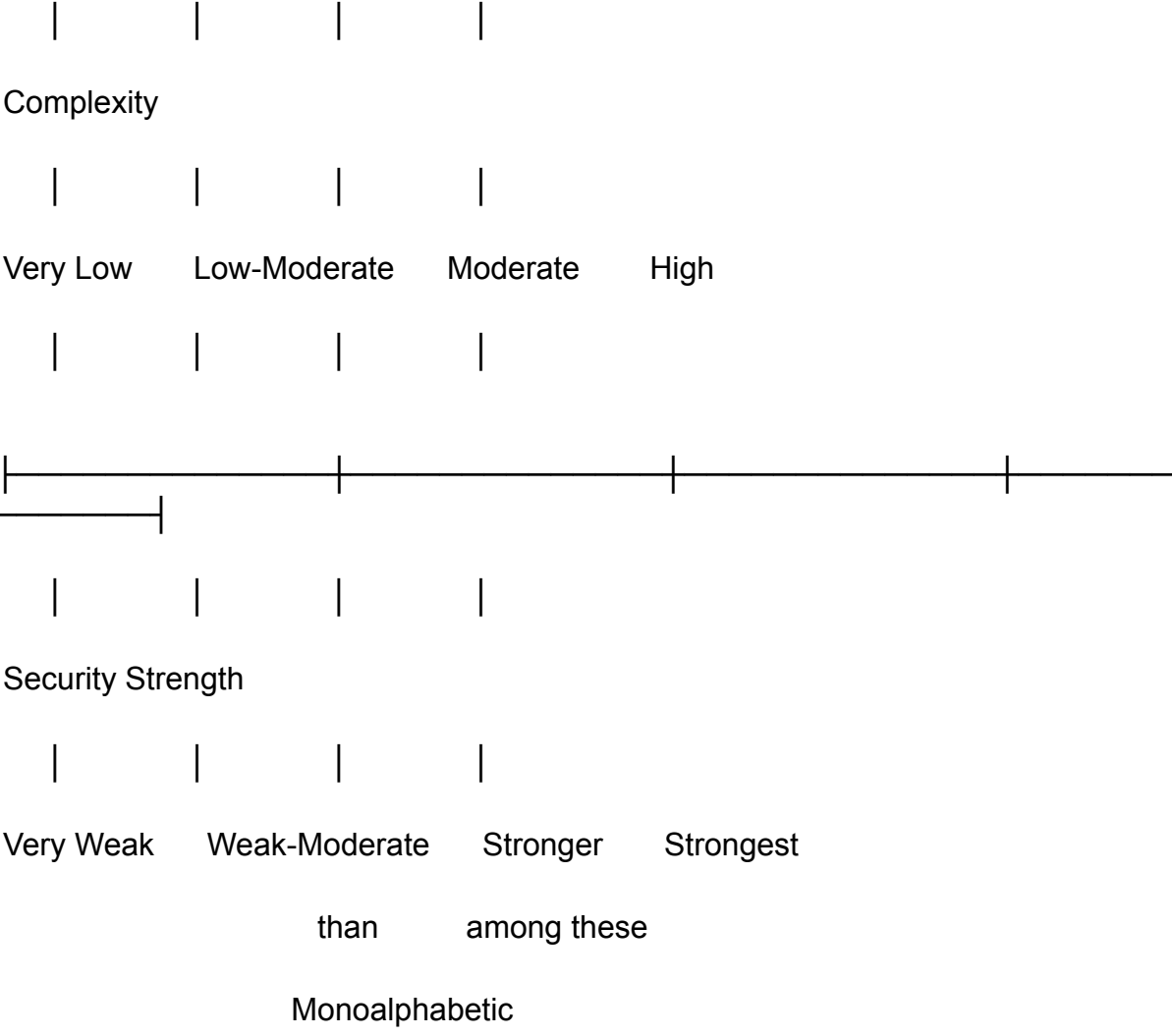
Signature of the student:

CONCEPT MAPPING

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

SUBSTITUTION TECHNIQUES





Technique	Key Usage		Complexity	Security Strength
Caesar Cipher	Single shift value		Very Low	Very Weak
Monoalphabetic Cipher	Random alphabet	substitution	Low	Weak to Moderate
Vigenère Cipher	Keyword repeated		Moderate	Strong
Hill Cipher	Matrix key		High	Very Strong

Analysis: Which Technique is Most Secure?

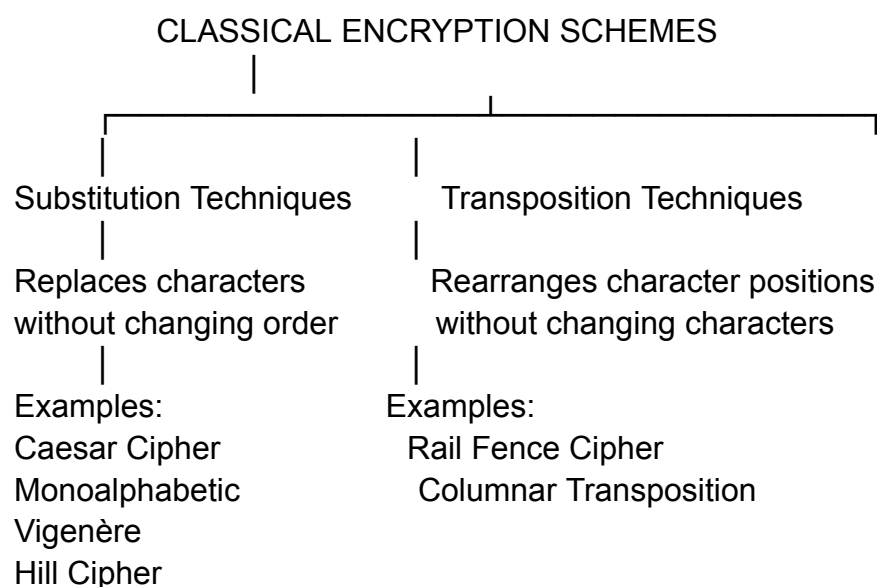
Hill Cipher is the most secure among these four classical substitution techniques because:

- Uses **matrix algebra** instead of simple letter replacement.
- Encrypts **multiple letters together (blocks)**.
- Produces greater diffusion, making frequency analysis much harder.
- A larger key space increases resistance against brute-force attacks.

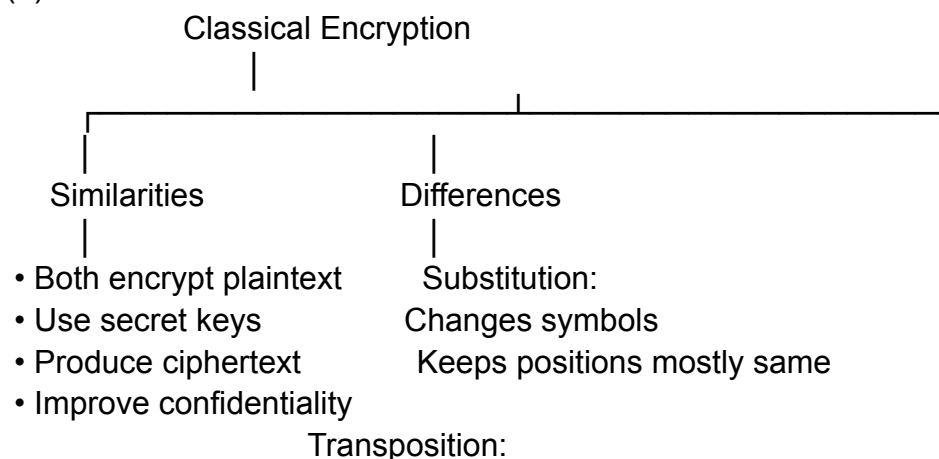
However, modern encryption algorithms (such as AES) are far more secure than all classical ciphers.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks: a. Show differences and similarities

- Map how each affects the Data structure and Security level
- Analyze which is more vulnerable to attacks and why



(a) Differences and Similarities



Changes positions only
Symbols remain unchanged

(i) Effect on Data Structure and Security Level

Feature	Substitution Techniques	Transposition Techniques
Data Structure	Characters are replaced	Character positions are rearranged
Character Frequency	Preserved	Preserved
Letter Positions	Mostly unchanged	Completely changed
Security Level	Moderate (depends on algorithm)	Moderate
Combined Use	Stronger security	Stronger when combined with substitution

(ii) Which is More Vulnerable to Attacks and Why?

Substitution Techniques

Vulnerable to:

- Frequency analysis
- Known-plaintext attacks
- Brute-force attacks (especially Caesar Cipher)

Reason:

Letter frequencies remain similar to the original language, making statistical attacks effective.

Transposition Techniques

Vulnerable to:

- Pattern analysis
- Anagramming
- Known-plaintext attacks

Reason:

The original letters are preserved, only their positions change. If enough ciphertext is available, attackers can reconstruct the original arrangement.

Overall Analysis

- **Caesar Cipher** is the weakest because it has only **25 possible keys**.
- **Monoalphabetic Cipher** has a much larger key space but is still vulnerable to frequency analysis.

- **Vigenère Cipher** improves security by using multiple substitution alphabets, reducing simple frequency-analysis effectiveness.
- **Hill Cipher** provides the highest security among the listed substitution techniques because it encrypts blocks of letters using matrix operations, making statistical attacks significantly more difficult.
- **Substitution** and **transposition** techniques each have weaknesses when used alone. Combining both methods (as done in many modern encryption designs) offers much stronger protection than using either technique by itself.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/techno	Good understanding with proper integration of most	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts

		logics	concepts		
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation